Supplemental Material for "Finite-size decay-mode hybridization near a mean-field fold in an open Rydberg ensemble"

OpenQuantum

I. GENERATOR, SYMMETRY SECTOR, AND PRECISION

We use $\gamma = 1$, $\Omega = 2$, and $V = 10$ unless stated otherwise. A permutation-invariant operator on N spin-1/2 particles has the form

$$\rho = \bigoplus_j \frac{\widetilde{\rho}_j}{d_j} \otimes I_{d_j}, \qquad d_j = \binom{N}{N/2-j} - \binom{N}{N/2-j-1}, \tag{S1}$$

where a binomial coefficient with lower argument -1 is zero. The stored PIQS blocks $\widetilde{\rho}_j$ sum over representation multiplicities. The trace is $\sum_j \operatorname{Tr}\widetilde{\rho}_j$, and the number of stored entries is $\sum_j (2j+1)^2 = \binom{N+3}{3}$. Local decay mixes different j blocks. All spectral and dynamical claims in the paper refer to this complete permutation-invariant operator space, not only $j = N/2$ and not to other permutation-symmetry sectors.

The Hamiltonian commutator is formed by column-major vectorization, $-i(I \otimes H - H^T \otimes I)$, followed by restriction to the Dicke blocks. The independent-emission coefficients follow the PIQS formulas [1], evaluated in double precision in `biorthogonal_generator.py`. For a target entry (j, m, m'), the diagonal decay coefficient is $-[N + (m + m')]/2$. Its off-diagonal sources have labels $(a, m_s, m'_s) = (j + \delta j, m + 1, m' + 1)$, with $\delta j = 0, +1, -1$. Their coefficients are, respectively,

$$\begin{aligned} &\frac{1}{2}\sqrt{(a+m_s)(a-m_s+1)(a+m'_s)(a-m'_s+1)}\,\frac{N/2+1}{a(a+1)}, \\ &\frac{1}{2}\sqrt{(a+m_s)(a+m_s-1)(a+m'_s)(a+m'_s-1)}\,\frac{N/2+a+1}{a(2a+1)}, \\ &\frac{1}{2}\sqrt{(a-m_s+1)(a-m_s+2)(a-m'_s+1)(a-m'_s+2)}\,\frac{N/2-a}{(a+1)(2a+1)}. \end{aligned} \tag{S2}$$

Only allowed source labels contribute; the first two expressions vanish when their denominators refer to an unavailable zero-spin transition. These coefficients are expressed in the multiplicity-summed storage convention.

An independent small-N check starts from the 2^N-dimensional spin Hilbert space with individual jump operators, and projects the full Liouvillian onto orthonormal symmetrized Pauli strings. The maximal matched spectral differences are shown in Table S1. This verifies the generator separately from eigensolver residuals, which cannot detect a common error in the matrix itself.

TABLE S1. Independent full-spin versus Dicke-block generator check. The invariance residual tests the full-spin action on the symmetrized Pauli basis.

N	Dimension	Maximum spectral difference	Invariance residual
1	4	1.99×10^{-15}	1.02×10^{-15}
2	10	1.01×10^{-14}	1.57×10^{-15}
3	20	3.77×10^{-14}	6.01×10^{-15}
4	35	4.46×10^{-14}	2.33×10^{-14}

This precision choice matters for a small gap. In the earlier PIQS-backed implementation used for exploratory calculations, internal emission coefficients led to a trace defect of approximately 4.7×10^{-7} at $N = 56$. The double-precision implementation reduces this to approximately 7.1×10^{-15} and changes that half-gap from 0.0003260261 to 0.0003309798. The finite-window gap exponent changes only from 0.140624 to 0.140501, but all figures retained in the present paper use the double-precision results. Earlier multi-drive quasipotential tests are not used as quantitative evidence in this manuscript.

II. FINITE-SIZE MODEL CONVENTIONS AND LITERATURE COMPARISONS

The locally damped Rydberg model in the main text is equivalent to those of Refs. [2, 3], after matching interaction and detuning conventions. With $\hat{N}_e = \hat{J}_z + N/2$ and $\hat{J}_\alpha = \frac{1}{2}\sum_i \hat{\sigma}_i^\alpha$, its Hamiltonian becomes, up to a scalar,

$$H = \Omega J_x + \frac{V}{2N} J_z^2 + \left[-\Delta + \frac{V(N-1)}{2N}\right] J_z. \tag{S3}$$

The exact finite-N correspondences are:

- Lee, Häffner, and Cross [2] use $V_{\mathrm{L}} \sum_{i<j} n_i n_j/(N-1)$, giving $V_{\mathrm{L}} = V(N-1)/N$, $\Delta_{\mathrm{L}} = \Delta$, and $\Omega_{\mathrm{L}} = \Omega$ for $N > 1$.
- In number-operator notation, Lee and Chan [3] use $V_{\mathrm{LC}} N_e^2/(2N) - \Delta_{\mathrm{LC}} N_e + \Omega_{\mathrm{LC}} J_x$. Thus $V_{\mathrm{LC}} = V$, $\Delta_{\mathrm{LC}} = \Delta + V/(2N)$, and $\Omega_{\mathrm{LC}} = \Omega$. Their collective Pauli operators are twice our J_α.
- Roberts and Clerk [4] use $J_{\mathrm{RC}} \sum_{i\neq j} S_i^z S_j^z + h^z J_z + h^x J_x$, with $S_i^z = \sigma_i^z/2$. Equation (S3) gives $J_{\mathrm{RC}} = V/(2N)$, $h^z = -\Delta + V(N-1)/(2N)$, and $h^x = \Omega$. Their collective-decay rate is set to zero; their exact steady state includes all spin blocks.
- For the Hamiltonian $\bar{J} J_z^2/N + \Delta_{\mathrm{NC}} J_z + \Omega_{\mathrm{NC}} J_x$ of Noachtar and Clerk [5], the mapping is $\bar{J} = V/2$, $\Delta_{\mathrm{NC}} = -\Delta + V(N-1)/(2N)$, and $\Omega_{\mathrm{NC}} = \Omega$. The coefficient $\bar{\Gamma}/N$ of $\mathcal{D}[J_-]$ is set to zero.

The local decay rate in every mapping is γ. The $O(1/N)$ detuning terms matter for a finite-size comparison, even though they vanish in the factorized limit.

Xiang et al. [6] instead use a heuristic jump operator within the maximum-spin Hilbert space $S = N/2$,

$$L_{\mathrm{eff}} = \sum_{M=-S+1}^{S} \sqrt{M+S}\, |S, M-1\rangle\langle S, M|. \tag{S4}$$

Their construction neglects phase differences from individual atomic jumps. It does not reproduce all inter-j transfers of $\sum_i \mathcal{D}[\sigma_i^-]$, which our generator retains. This distinction concerns the finite-N dissipator, independently of any shared mean-field equations.

III. MEAN-FIELD JACOBIAN AND INTERPRETATION OF THE FOLD

For the Bloch coordinates $q = (n, u, v)$ of the main text, the Jacobian is

$$J = \begin{pmatrix} -1 & 0 & -\Omega \\ Vv & -1/2 & -(\Delta - Vn) \\ \Omega - Vu & \Delta - Vn & -1/2 \end{pmatrix}. \tag{S5}$$

At stationarity, $v = -n/\Omega$ and $u = 2n(\Delta - Vn)/\Omega$. Substituting these relations into $\det(\lambda I - J)$, and reducing its difference from $\lambda[(\lambda+1)^2 + \Omega^2/2]$ by the polynomial constraints $F = F_n = 0$, gives zero identically. This is implemented symbolically in `pra_meanfield.py` and also checked numerically at both folds. For the parameters of the paper, the right fold is $(\Delta, n) = (4.5749637137, 0.4305200135)$; both folds have Jacobian eigenvalues $0, -1 \pm i\sqrt{2}$ to numerical precision.

The simple zero eigenvalue describes the linearization at the coalescing fixed points. The stationary cubic has a higher-order root at the cusp, without changing the zero root's algebraic multiplicity in this Jacobian. The stationary-root exponents of Li, Sun, and Yi [7] concern the former polynomial. Their mean-field conventions map to ours through $\Omega_{\mathrm{CPB}} = \Omega/2$, $\Delta_{\mathrm{CPB}} = -\Delta$, $V_{\mathrm{CPB}} = V$, and $\Gamma = \gamma$. This mapping does not identify their nonlinear stationary-root structure with a defective Jacobian or a finite-N many-body EP.

The separate exclusion of nontrivial Jordan blocks at zero follows from bounded evolution under a finite-dimensional completely positive and trace-preserving (CPTP) semigroup [8]. It applies to the physical real-parameter generator. The complex-detuning continuation below is generally not CPTP, so that argument does not constrain its Jordan structure.

A. Reference values for the real-axis crossings

The left-fold coordinates are $(\Delta_f, n_f) = (3.5996368905, 0.1569270404)$. For $N = 24, 28, \ldots, 56$, the real-mode crossing moves from $\Delta_*(24) = 3.13740295$ to $\Delta_*(56) = 3.30652148$, while its half-gap decreases from 0.0292933152 to 0.000330979777. Fits to $\ln g_N$ give $\sigma_{24:56} = 0.14050$ and $\sigma_{40:56} = 0.14305$; successive four-atom slopes increase from 0.13218 to 0.14383. These window-dependent slopes do not determine an asymptotic exponent or convergence to the fold.

At $N = 32$ and $\Delta_* = 3.205747775$, representative full-generator eigenvalues are

$$\lambda_- = -0.765098904, \qquad \lambda_+ = -0.745092273, \tag{S6}$$

$$\lambda_{\rm osc} = -0.725036287 \pm 2.864794973i. \tag{S7}$$

The oscillatory pair decays more slowly than both target real modes. The ordinary Liouvillian gap therefore differs from the real-pair half-gap even though the latter becomes small. Full-precision crossing records are retained in `data/biorthogonal_audit/`.

IV. CONSTRUCTION AND VALIDATION OF PHYSICAL PREPARATIONS

Let a strictly positive single-atom density matrix have Bloch vector $\boldsymbol{b}$, with $r = |\boldsymbol{b}| < 1$. In the spin convention used by PIQS, the coordinates of the main text give

$$\boldsymbol{b} = (2u, -2v, 2n-1), \qquad \rho_1 = \sqrt{\det \rho_1}\, \exp[\operatorname{arctanh}(r)\, \widehat{\boldsymbol{b}} \cdot \boldsymbol{\sigma}]. \tag{S8}$$

The multiplicity-summed blocks of its product state are therefore

$$\widetilde{\rho}_j = d_j (\det \rho_1)^{N/2} \exp[2 \operatorname{arctanh}(r)\, \widehat{\boldsymbol{b}} \cdot \boldsymbol{J}_j]. \tag{S9}$$

We diagonalize the Hermitian exponent within each block and evaluate the scalar weights in logarithmic form. This avoids constructing a $2^N \times 2^N$ density matrix. The trace is checked before normalization.

For $N = 1, 2, 4$, the resulting excitation-number distribution agrees with $\binom{N}{k} n^k (1-n)^{N-k}$ to less than 1.7×10^{-16}, and the reconstructed single-atom coherences agree to less than 1.2×10^{-16}. Positivity follows directly from Eq. (S9); the single-atom states used in the calculations are strictly inside the Bloch ball. A separate $N = 1$–4 test starts from explicit tensor products and individual spin jumps, and integrates the full-spin equation with an adaptive DOP853 solver. At $t = 0, 0.25, 0.75, 1.5$, its excitation density and purity agree with Dicke-block exponential multiplication to less than 2×10^{-12}. This checks the preparation and physical trace convention in time evolution, independently of the spectral comparison in Table S1.

The high-branch difference is

$$\delta\rho_{\rm hi} = \rho_1(q_{\rm hi} + \eta e_{\rm hi})^{\otimes N} - \rho_1(q_{\rm hi} - \eta e_{\rm hi})^{\otimes N}, \tag{S10}$$

where $q_{\rm hi}$ is the surviving high mean-field fixed point, $e_{\rm hi}$ is its Euclidean-normalized real relaxation eigenvector with positive population component, and $\eta = 0.005$. Both $q_{\rm hi}$ and $e_{\rm hi}$ are evaluated at the current detuning in the original mode-character scan. The later fixed-preparation controls instead freeze these values at $\Delta_*(N)$. The second difference is

$$\delta\rho_f = \rho_1(q_f)^{\otimes N} - \rho_{\rm ss}(\Delta), \qquad q_f = (0.15692704, 0.31861940, -0.07846352). \tag{S11}$$

The fold state is fixed while Δ is varied. It is not asserted to be a stationary or metastable state of the finite generator at those detunings.

For each detuning the real spectral pair is computed together with its transpose-left dual. If $H = W_d^T L R_d$ is the diabatic representation, diagonalizing $H = T\Lambda T^{-1}$ gives eigenmode bases $R = R_d T$ and $W = W_d (T^{-1})^T$. The observable residues are calculated from these exact eigenmode bases, not from the diagonal entries of the diabatic matrix. Scanning 15 detunings over $\Delta_*(32) \pm 0.06$ resolves the transfer of response; the endpoints and crossing are separately evaluated for $N = 40$. The largest right invariant-subspace residual in the $N = 32$ scan is 1.46×10^{-12}.

At $N = 32$, changing the ARPACK seed and shift from -0.76 to -0.80 changes the two real eigenvalues at the three diagnostic detunings by less than 9.0×10^{-15}. Halving η changes the residue vector divided by 2η by 4.96×10^{-6}, 7.29×10^{-6}, and 5.75×10^{-6} in relative Euclidean norm, respectively. These checks support the perturbative interpretation of the first preparation. They do not turn the second, finite displacement into a linearized mean-field trajectory.

For reference, the $N = 32$ high-branch residues (a_-, a_+) at $\delta = \mp 0.06$ are $(8.94 \times 10^{-6}, 1.1694 \times 10^{-3})$ and $(1.1082 \times 10^{-3}, -2.91 \times 10^{-6})$, respectively. The corresponding low-density absolute-profile fractions, using $k/N < 0.22$, are $(0.209, 0.0305)$ and $(0.0314, 0.237)$. At $N = 40$, the nondominant high-branch residue at either endpoint is below 7.3×10^{-7}. At the crossing center, the fold-state residues are approximately $(-1.666, 1.423)$ for $N = 32$ and $(-5.475, 5.217)$ for $N = 40$. These values characterize signed eigenmode weights; they are not state probabilities.

V. BIORTHOGONAL PROJECTION AND NORMALIZATION CONVENTIONS

The left eigenvectors are computed from L^T. In column-major vectorization the bilinear pairing $W^T R$ is the algebraic dual; physical observables use the corresponding transpose. For each simple real decay mode, we phase-align and Hermitize its right and dual left eigenoperators. We then enforce $W^T R = I$ in the selected two-dimensional subspace. Eigensolver, Hermiticity, trace, and subspace residuals are retained separately. Observable residues involve both excitation and detection overlaps; small left-right overlaps can produce large expansion coefficients [9].

For the two real modes, the combined density response can be written exactly as

$$a_+ e^{(\bar{\lambda}+g)t} + a_- e^{(\bar{\lambda}-g)t} = e^{\bar{\lambda}t} \left[(a_+ + a_-) \cosh(gt) + (a_+ - a_-) \sinh(gt) \right]. \tag{S12}$$

If, along a limiting sequence, $g \to 0$, $\bar{\lambda} \to \lambda_0$, $a_+ + a_- \to A$, and $g(a_+ - a_-) \to B$, the fixed-time limit is $(A + Bt)e^{\lambda_0 t}$. Molina [10] derives the corresponding opposite-residue structure for a collective-spin exceptional spectral phase. Our finite-N values demonstrate cancellation, but do not establish these limits or a real-axis Jordan block. Eigenmode residues are not probabilities of metastable phases.

In this basis the derivative matrix $W^T L_1 R$ has two distinct real eigenvalues. Diagonalizing it defines the local diabatic modes. At the projection point, the derivative of the two-eigenvalue discriminant is unaffected by basis-derivative commutators. This gives the stationarity identity used in the main text. Numerically, the minimum is refined using

$$\delta\Delta = -\frac{e_a - e_b}{s_a - s_b} \tag{S13}$$

until $|\delta\Delta| < 10^{-11}$, with independently checked eigenmode and subspace residuals.

Let $P = R_d W_d^T$ be the generally oblique spectral projector at Δ_* and $Q = I - P$. Applying the spectral-parameter-dependent projection construction [11, 12], with $[P, L_*] = 0$ at $L_* = L(\Delta_*)$, gives

$$\begin{aligned} H_{\text{eff}}(z, \delta) &= H_P(\delta) \\ &\quad + \delta^2 W_d^T L_1 Q [z - Q(L_* + \delta L_1) Q]_Q^{-1} Q L_1 R_d. \end{aligned} \tag{S14}$$

The inverse acts on the range of Q. Thus the frozen linear matrix reproduces the value and first derivative at the reference point but not the complete neighboring spectrum. At $N = 40$, over $|\delta| \le 2g/|s_a - s_b|$, its largest tested half-gap relative error is 1.39×10^{-3}. A small absolute correction is not automatically negligible relative to a much smaller off-diagonal element.

The population-peak convention fixes $\max_k |P_i(k)| = 1$ for $P_i(k) = \mathrm{Tr}(\Pi_k R_i)$, with the largest-magnitude entry positive. In the stored convention of Eq. (S1), the physical trace norm of a Hermitian right operator is

$$\|R\|_1 = \sum_j \mathrm{Tr}\, |\widetilde{R}_j|. \tag{S15}$$

The corresponding norm of a dual observable is the maximum operator norm over its blocks. No extra representation-multiplicity factor is inserted on the left. Table S2 lists descriptive logarithmic slopes for the same data window under different normalizations. The balanced normalization uses $c_b/c_a = \sqrt{|V_{ba}/V_{ab}|}$ and forces equal magnitudes.

TABLE S2. Finite-window coupling exponents for $N = 24, 28, \ldots, 56$. The individual values are normalization-dependent.

Right-mode normalization	α_{ab}	α_{ba}	$(\alpha_{ab} + \alpha_{ba})/2$
Population peak	0.00963768	0.27136532	0.14050150
Physical trace norm	0.00356277	0.27744023	0.14050150
Balanced	0.14050150	0.14050150	0.14050150

VI. BOUNDARY FLUX AND THE LIMITS OF TAIL ARGUMENTS

For a diagonal mask χ in the stored operator coordinates, define

$$J_{ab} = W_a^T[\chi, L]R_b, \qquad D_a = W_a^T\chi R_a, \qquad O_{ab} = W_a^T\chi R_b. \tag{S16}$$

The invariance equations $LR = RH$ and $W^TL = HW^T$ give

$$J_{ab} = (e_b - e_a)O_{ab} + V_{ab}(D_a - D_b). \tag{S17}$$

At the gap minimum, $e_a = e_b$, this is a flux representation of the cross coupling. The masks used in the diagnostic are

$$\chi_{(j,m,m')} = \mathbf{1}\left[\frac{k_m + k_{m'}}{2N} \le n_c\right], \qquad k_m = m + N/2. \tag{S18}$$

They are coordinate masks, not quantum channels or probabilities of measurement outcomes.

Eight cuts and nine sizes reconstruct the cross couplings with relative error below 1.6×10^{-8}. Nevertheless, splitting the weak flux into an absolute envelope and a cancellation factor is not invariant. For $n_c = 0.157$, their fitted exponents over $N = 24$–56 are approximately 0.13123 and 0.13942; at $n_c = 0.30$ they become 0.01137 and 0.25778. The denominator correction restores the same total coupling exponent. This prevents interpreting the two contributions at one selected cut as two independently determined activation actions.

At $N = 40$, the output-diagonal and output-off-diagonal contributions to $W_b^TLR_a$ in the population-peak convention are 0.9659542705 and -0.9657905699. Their sum is $V_{ba} \simeq 1.6370 \times 10^{-4}$. The displayed terms are rounded; this coordinate partition does not define two independent physical processes.

The weak-coupling contraction can also be written $W_b^T(L - e_a)R_a$. In the exact projected subspace, $(L - e_a)R_a = V_{ba}R_b$, so the small quantity is transferred from a large cancelling sum into an exponentially small residual. An asymptotic calculation must determine that residual independently; using exact spectral modes to reconstruct it is not a WKB prediction.

VII. COMPLEX-DETUNING EXCEPTIONAL POINTS

We use a full-generator complex continuation $L(\zeta) = L_0 + \zeta L_1$, independently of the real spectral-parameter recursion. A two-sided block inverse iteration, with QR orthonormalization, gives right and transpose-left invariant subspaces. After dual normalization, define

$$H = W^TLR, \quad B = W^TL_1R, \quad D = \left(\frac{H_{00} - H_{11}}{2}\right)^2 + H_{01}H_{10}. \tag{S19}$$

Because D is a similarity invariant, its derivative is

$$D' = \frac{(H_{00} - H_{11})(B_{00} - B_{11})}{2} + B_{01}H_{10} + H_{01}B_{10}. \tag{S20}$$

Newton iterations update both the detuning and invariant subspaces; they do not solve only a frozen two-by-two model. The QR basis remains regular at an EP, where individual eigenvectors become ill-conditioned.

TABLE S3. Double-precision complex-detuning EPs. $M = H - \frac{1}{2}\,\mathrm{Tr}(H)I$. Rates are in units of γ. The norm of M is evaluated with an orthonormal right-subspace basis.

N	ζ_{EP}	λ_{EP}	$\|M\|_F$	$\|M^2\|_F/\|M\|_F^2$
24	$3.136806063 + 0.026370766i$	$-0.751013173 + 0.031608734i$	0.25945	2.80×10^{-15}
32	$3.205686613 + 0.007464558i$	$-0.755104588 + 0.010634053i$	0.26380	2.10×10^{-15}
40	$3.250218904 + 0.002129005i$	$-0.757677321 + 0.003436485i$	0.26840	3.24×10^{-15}

The largest left or right subspace residual at the final EP points is 1.61×10^{-12}. Independent full-matrix ARPACK candidates have target-mode residuals below 1.68×10^{-13}. Direct eigenvalues at the EP still split by roughly 4–7×10^{-8} in floating-point arithmetic; we do not report this as an exact zero. The nonzero nilpotent part, simple zero of D, and

the branch exchange provide complementary local evidence. Eight nearby candidate modes are retained; the nearest sampled external-mode distance is 0.457–0.460. This is not a global exclusion of other EPs or other spectral sectors.

For $N = 32$, we take $\zeta(\theta) = \zeta_{\mathrm{EP}} + re^{i\theta}$ with $r = 0.00223936747$, less than the distance to the conjugate EP. At 24 points per circuit, the one-circuit exchange and two-circuit return errors are 9.75×10^{-14} and 8.41×10^{-14}. The discriminant winds once and twice, respectively. These tests concern the square-root eigenvalue branches [13]; a two-circuit eigenvalue return does not determine the phase holonomy of normalized eigenvectors. Repeating with a different random seed, inverse-iteration shift, initial detuning, and 48 points per circuit changes the EP by 4.83×10^{-15} and common-node eigenvalues by at most 2.24×10^{-13}. A complex-direction central difference of D agrees with D' to relative error 4.10×10^{-8}.

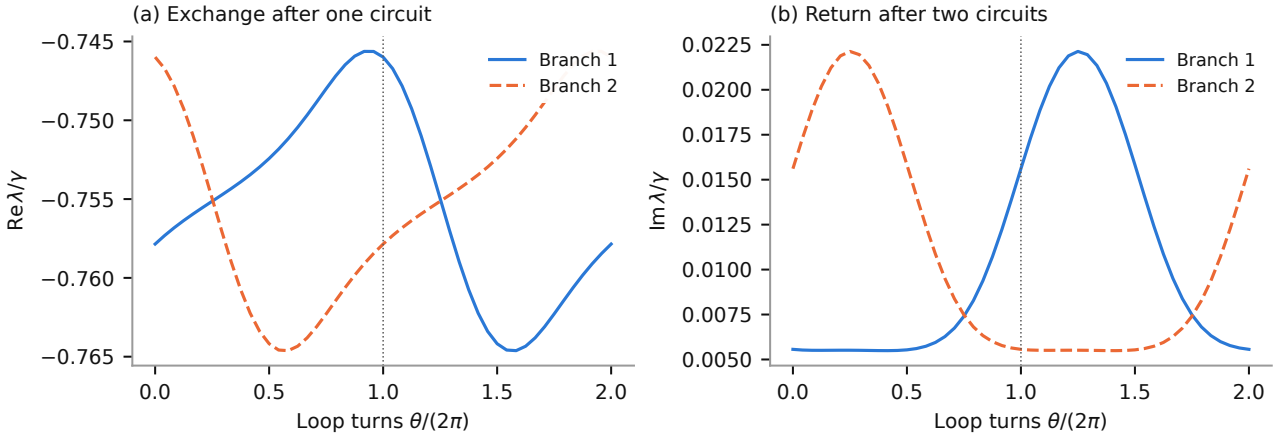


FIG. S1. Full-generator eigenvalue continuation around the $N = 32$ complex-detuning EP. Branch labels are followed continuously, rather than resorted at each point. The two branches exchange after one circuit and return after two. This complex-parameter loop is a spectral diagnostic, not a physical CPTP evolution.

VIII. COHERENCE RECURSION, DERIVATIVES, AND REAL-ROOT VALIDATION

The recursion in the main text is a matrix continued fraction [14] evaluating a Schur effective operator [12]. It retains spectral dependence exactly where the eliminated block and intermediate denominators are invertible. No adiabatic or Markov approximation is introduced by this algebraic elimination.

For real z, each denominator $X_q = zI - A_q - \Sigma_q(z)$ is inverted by a linear solve. Set $G_q = X_q^{-1}$ and $E_q = I - \Sigma_q'$. Differentiation of the positive-order recursion gives

$$\Sigma'_{q-1} = -U_{q-1}G_qE_qG_qD_{q-1}, \tag{S21}$$

$$\Sigma''_{q-1} = U_{q-1}\left[2G_qE_qG_qE_qG_q + G_q\Sigma''_qG_q\right]D_{q-1}. \tag{S22}$$

With $K = A_0 + 2\,\mathrm{Re}\,\Sigma_0$, a simple real eigenbranch obeys

$$\mu' = l^TK'r, \qquad l^Tr = 1, \tag{S23}$$

$$(K - \mu I)r' = -(K' - \mu' I)r, \qquad l^Tr' = 0, \tag{S24}$$

$$\mu'' = l^TK''r + 2l^TK'r'. \tag{S25}$$

The middle equation is solved as a bordered linear system. All derivatives are with respect to the spectral variable z, not detuning. At the four minima $|f'| < 2.3 \times 10^{-12}$; eigenmode and differentiated-equation residuals are below 5.1×10^{-13}. For $N = 24$, central differences with step 10^{-4} agree with the first and second operator derivatives to relative errors 3.65×10^{-8} and 1.53×10^{-7}.

The root search selects the nearest effective real eigenvalue only within the local window. Such a selection is not a global branch-tracking algorithm and can jump between branches. A final residual check is therefore mandatory. In particular, $K = \mathrm{diag}(-0.80, -0.72)$ has no self-consistent root in $[-0.78, -0.74]$, although nearest-branch selection changes sign there and can make Brent's method report convergence. The implementation rejects this false root

because the final residual is about 0.04. It also rejects a nearest effective eigenvalue with any nonzero imaginary part returned by the real-matrix eigensolver; a small complex pair is not rounded into a real root. Complex spectral inputs or detunings are rejected by the $2\,\mathrm{Re}\,\Sigma$ implementation.

For all eight retained roots, back-substitution reconstructs the complete right eigenoperator. The largest $\|LR - zR\|_2/\|R\|_2$ is 1.13×10^{-13} and the largest normalized trace is 8.2×10^{-14}. The $q = 0$ component carries between 0.360 and 0.419 of the full Euclidean norm. The smallest one-norm reciprocal condition estimates of the intermediate denominators at the roots range from 4.33×10^{-4} to 1.73×10^{-3}. These checks test the computed points, not absence of poles over an entire real or complex domain. Small residuals alone are not rigorous forward-error bounds for a nonnormal matrix; the agreement with independent full-generator spectra is also retained.

A. Minimum depth, curvature, and quadratic gap

Table S4 gives the local minimum parameters and the error of $g_{N,\mathrm{quad}} = \sqrt{2\epsilon_N/c_N}$. Direct roots of $f_N = 0$ instead reproduce the independent full-generator half-gaps to relative differences below 1.3×10^{-9}. This root-reconstruction accuracy is distinct from the larger error of the quadratic expansion.

TABLE S4. Local real-axis coherence reduction. g_N is the independently computed full-generator half-gap; the last column is $g_{N,\mathrm{quad}}/g_N - 1$. All entries use $\gamma = 1$.

N	ϵ_N	c_N	g_N	Relative error
24	9.33404×10^{-3}	22.28695	0.02929332	-1.2002×10^{-2}
32	1.09619×10^{-3}	21.96534	0.01000332	-1.2773×10^{-3}
40	1.28422×10^{-4}	24.11955	0.00326360	-1.0842×10^{-4}
48	1.51689×10^{-5}	27.82335	0.00104421	-6.5891×10^{-6}

The minimum need not coincide with the root midpoint: at $N = 24$, $z_c = -0.745502759$, whereas $(z_- + z_+)/2 = -0.750910792$. For $N = 48$, the nearest eigenvalues of the frozen $K_N(z_c)$ include -0.75910208, -0.21846755, and zero; the two full-generator roots are -0.76013902 and -0.75805059. Thus freezing this effective operator removes the spectral dependence that retains the nearby pair.

Over the four sizes in Table S4, a fit of $\ln \epsilon_N$ gives $\alpha_\epsilon = 0.267635$. The same convention applied to the curvature gives $\alpha_c = -0.009490$, so the fitted quadratic-gap exponent is $(\alpha_\epsilon - \alpha_c)/2 = 0.138562$, compared with 0.139030 for the exact gap on the same window. This four-size window differs from the nine-size crossing fit. Neither fit establishes an asymptotic action or an algebraic curvature prefactor.

B. Derivative pairing and elimination poles

For $T(z) = zI - K(z)$ at a simple nonlinear eigenvalue λ, the singular part of the reduced resolvent is

$$T(z)^{-1} = \frac{r l^T}{(z - \lambda)\, l^T T'(\lambda) r} + O(1), \qquad T'(\lambda) = I - K'(\lambda). \tag{S26}$$

This derivative normalization is standard in nonlinear eigenvalue theory [15] and appears in Liouvillian projection methods [12]. With the main text's back-substitution, its denominator equals the full left-right pairing. At an effective eigenbranch's off-root minimum, a vanishing derivative pairing does not imply a pole, because $T(z_c)$ still has no zero eigenvalue on that branch.

If elimination remains regular, the effective branch is algebraically simple, and $f = f' = 0$ with $f'' \neq 0$, the Schur determinant has a double zero. Back-substitution preserves the one-dimensional kernel, producing the usual EP2 structure [13]. At a pole of the eliminated resolvent these arguments do not apply without examining the full block matrix. Neither this conditional criterion nor a local quadratic fit determines the large-N minimum depth.

C. Initial-coherence source and physical residues

For the block decomposition $L = \begin{pmatrix} L_{00} & B \\ C & L_{QQ} \end{pmatrix}$ and an initial operator x, the Laplace-domain equation $(zI - L)\widetilde{x} = x$ gives

$$\widetilde{x}_Q = G(z)[x_Q + C\widetilde{x}_0], \tag{S27}$$

$$[zI - K(z)]\widetilde{x}_0 = b(z), \qquad b(z) = x_0 + BG(z)x_Q. \tag{S28}$$

The Laplace transform is initially defined in its convergence half-plane; residues at negative decay rates use its meromorphic continuation. Applying the simple-root expansion above to a population observable yields

$$a_i = \frac{(o_0^T r_i)\, l_i^T [x_0 + BG(\lambda_i)x_Q]}{l_i^T [I - K'(\lambda_i)] r_i}. \tag{S29}$$

Both the source and denominator are essential. This is an application of the existing nonlinear resolvent expansion [12, 15], not a new general residue theorem. The assumptions are a simple full root, a simple effective branch, and regular elimination near that root. For an observable with a nonzero Q component, the detection factor must also include $o_Q^T GCr_i$; Eq. (S29) uses $o_Q = 0$.

The source is evaluated by the same positive-coherence recursion as the effective operator. With $X_q = zI - A_q - \Sigma_q(z)$, define

$$\beta_{q_{\mathrm{top}}} = x_{q_{\mathrm{top}}}, \qquad \beta_q = x_q + U_q X_{q+1}^{-1} \beta_{q+1} \quad (q \geq 1). \tag{S30}$$

For real z and Hermitian x, $b(z) = x_0 + 2\,\mathrm{Re}[U_0 X_1^{-1} \beta_1]$. This conjugate-sector formula is not a complex-parameter continuation. Right and transpose-left modes are separately back-substituted to check the source contraction and derivative denominator. The largest full-mode residual over the eight roots is 1.14×10^{-13}; the derivative pairing agrees with $W^T R$ to relative error below 9.3×10^{-14}. At $N = 4$, $z = 0.37$, a direct dense full-resolvent solve agrees with the reduced source equation to relative error 4.95×10^{-16}.

We use the previously saved root positions, without refitting them, and compare with independently stored full-generator mode arrays. The sources are the high-branch difference, the fold product state itself, and the coherence-only difference defined below. A stationary subtraction changes no nonzero-mode residue because $W_i^T \rho_{\mathrm{ss}} = 0$ for $\lambda_i \neq 0$. However, this equivalence does not survive an inconsistent source ablation; all fold-source ablations below use the unsubtracted product state. The normalized error is $\|a_{\mathrm{approximation}} - a_{\mathrm{full}}\|_2 / \|a_{\mathrm{full}}\|_2$, over the two real-mode residues.

TABLE S5. Schur-response validation at the crossing. The full-error column is the largest residue-vector error over the three sources. The other columns keep the roots fixed and remove only the stated response factor. The fold-source ablation uses ρ_f, without stationary subtraction.

N	Full error	High, omit BGx_Q	Fold, omit BGx_Q	High, replace denominator by 1
24	1.30×10^{-12}	3.694	0.7110	1.090
32	2.62×10^{-11}	10.898	0.7401	1.018
40	2.44×10^{-10}	29.913	0.7595	1.003
48	2.08×10^{-9}	80.462	0.7734	1.0005

For example, at $N = 32$ the high-branch residues are $(6.16285, 5.26413) \times 10^{-4}$. Omitting the source contribution instead gives $(7.07828, -5.49532) \times 10^{-3}$. The signs of the derivative denominators at the two roots are opposite, with values 0.193815 and -0.250672 under $l^T r = 1$. Replacing them by one yields $(1.19445, -1.31957) \times 10^{-4}$. These are finite-size response failures; the dimensionless denominator is not a probability or a decay rate.

D. Identical populations with different initial coherences

Let $\rho_f = \rho_1(q_f)^{\otimes N}$ and define

$$\rho_d = \sum_k \Pi_k \rho_f \Pi_k = \frac{1}{2\pi} \int_0^{2\pi} d\theta\, e^{-i\theta N_e} \rho_f e^{i\theta N_e}. \tag{S31}$$

This global phase average is a CPTP preparation. It removes $q \neq 0$ while preserving every $q = 0$ Dicke population; it is not independent local dephasing. Both states are positive and normalized. Their difference $\delta\rho_c = \rho_f - \rho_d$ has $x_0 = 0$ and zero initial population signal, but $x_Q \neq 0$. For the fixed drive phase used in the paper, the exact Heisenberg equation gives

$$\delta\dot{n}(0) = -\Omega v_f = n_f = 0.1569270404. \tag{S32}$$

Thus population-only initialization already fails at the first time derivative. For $N = 32$, sampling $t = 0, 0.1, \ldots, 12$ gives the largest magnitude $|\delta n| = 0.1343027290$ at $t = 1.1$. Its two real-mode residues are $(-1.2360093, 1.0491984)$, recovered by Eq. (S29); omitting BGx_Q makes both identically zero. The real-pair contribution alone has relative L^2 error 1.00853 over $6 \leq t \leq 12$, so this control establishes source dependence, not a two-mode dynamical closure. No new four-mode accuracy claim is made for this preparation.

An independent construction in the full 2^N spin Hilbert space verifies global dephasing and its density response for $N = 1$–4, with maximum density error 5.3×10^{-15} at five times up to $t = 2$. At $N = 32$, an adaptive DOP853 integration additionally checks the sparse-exponential time series, with maximum absolute density difference 6.9×10^{-14}. All source records, errors, and full signals are saved in `data/coherence_response/`. This finite-size calculation does not establish a $1/g_N$ residue asymptote or a real-axis Jordan limit.

E. Truncating coherence orders

Keeping only $|q| \leq q_{\max}$ defines a truncated matrix, not a CPTP approximation. At low cutoffs the nearest real pair can change identity; all candidate eigenvalues are saved. Representative relative half-gap errors are given in Table S6. Nonmonotonicity and different errors in the individual eigenvalues prevent treating one favorable gap value as convergence of the full spectrum.

TABLE S6. Relative half-gap error from coherence truncation. Full-generator values use all $|q| \leq N$.

$q_{\max}$	$N = 40$	$N = 56$
8	$+5.1873$	$+165.17$
10	$+0.35523$	$+24.376$
12	$+0.025262$	$+3.7778$
14	-1.704×10^{-4}	$+0.65009$
16	-8.49×10^{-5}	-2.28×10^{-5}
20	-1.45×10^{-5}	$+2.35 \times 10^{-7}$
24	-5.66×10^{-7}	-1.83×10^{-6}

IX. TIME-DOMAIN COMPARISON AND ERROR DEFINITIONS

The full evolution is evaluated by sparse exponential multiplication for 121 times $t = 0, 0.1, \ldots, 12$, acting directly on the two initial differences. This avoids a time-discretized master-equation integrator. Trace conservation of the evolved differences is checked; the largest error in the three diagnostic detunings is below 9×10^{-16}.

The four-mode approximation includes the two real modes, the oscillatory eigenmode found near $-0.72 + 2.85i$, and its Hermitian-conjugate partner. Both right and transpose-left eigenmode residuals are checked. The residues are projected from the chosen preparations and observable, without fitting any time-domain parameters. For a lower time cutoff t_0, the reported relative error is

$$E(t_0) = \frac{\left[\sum_{t\in[t_0,12]} |\delta n_{\text{approx}}(t) - \delta n_{\text{full}}(t)|^2\right]^{1/2}}{\left[\sum_{t\in[t_0,12]} |\delta n_{\text{full}}(t)|^2\right]^{1/2}}. \tag{S33}$$

The exponential envelope removed in Fig. S2 is not removed when calculating errors. Table S7 lists the $t_0 = 6$ comparisons. Earlier times generally contain larger contributions from omitted modes; for example, at the crossing the high-branch four-mode error is 1.49% over $4 \leq t \leq 12$.

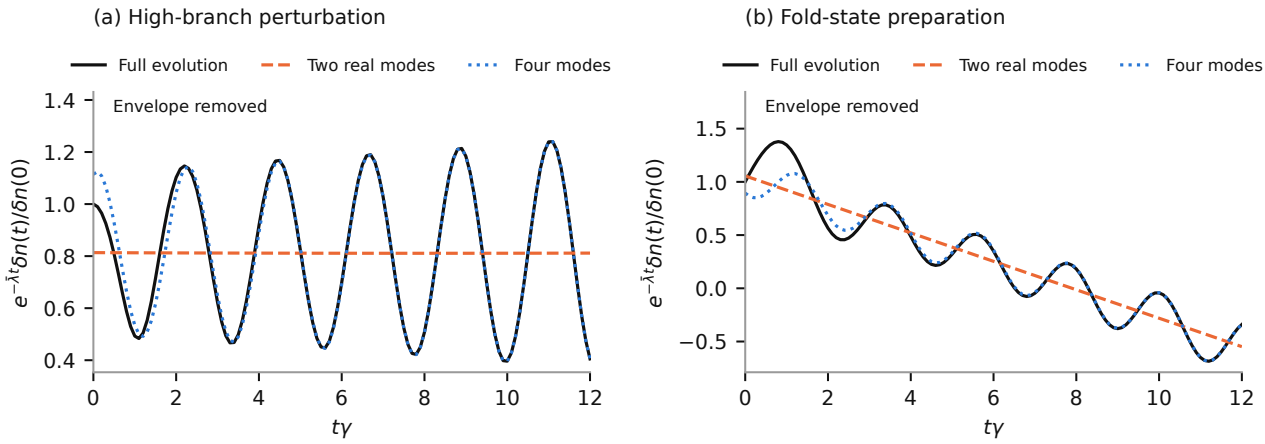


FIG. S2. Spectral contributions to the density transient at the $N = 32$ crossing. Solid black: complete permutation-invariant evolution. Orange dashed: the target real pair with exact residues. Blue dotted: those modes plus the oscillatory conjugate pair. (a) High-branch product-state difference; (b) fold product state relative to the stationary state. Each signal is divided by its initial value and multiplied by $e^{-\bar{\lambda}t}$, where $\bar{\lambda} = (\lambda_- + \lambda_+)/2$. This display exposes omitted oscillations; the errors and physical-effect tests use raw density signals.

TABLE S7. Relative L^2 errors over $6 \le t \le 12$ for $N = 32$. All values are percentages.

$\Delta - \Delta_*$	High, two real	High, four modes	Fold state, two real	Fold state, four modes
-0.06	25.14	0.481	46.34	5.14
0	25.10	0.506	69.47	5.00
$+0.06$	25.78	0.529	4.99	0.256

These are finite-time, preparation-dependent tests. The oscillatory pair has the less negative real part at the crossing, so the target real pair's small separation does not justify omitting it. Four modes do not close the dynamics exactly or uniformly over all initial states. More generally, large expansion coefficients can extend transients beyond an inverse-gap estimate [9, 16]. Our reported errors test the specified preparations and time interval, without establishing or excluding such behavior for every physical initial state.

X. FIXED-PREPARATION CONTROLS FOR THE COMPLETE DENSITY RESPONSE

A. Preparations, baselines, and size control

The two sources are $\delta\rho_c = \rho_f - \rho_d$ and the high-branch difference frozen at $\Delta_*(N)$, denoted by $\delta\rho^*_{\rm hi}$. Neither changes during a detuning scan. The high-branch seed depends on N; the fold single-atom seed is common to all sizes. We use $N = 24, 32, 40, 48$, with $\delta = 0, \pm w_N, \pm 4w_N, \pm 0.06$ and $w_N = 2g_N/|s_a - s_b|$. These are seven distinct points per size; their ordering by distance changes with N. Each full signal is sampled at $t = 0, 0.1, \ldots, 12$. All reported maxima refer to this grid. Side-point residues use newly computed full-generator subspaces, not a frozen two-by-two projection.

For $V = 0$, the density response follows the exact one-atom Bloch equation at the same bare detuning. This remains true for the correlated common-phase mixture, because density is a one-body observable. A separate initial-Hartree-matched reference uses $\Delta_{\rm free} = \Delta - V(N-1)n_{\rm seed}/N$. At $\Delta_*(32)$ the coherence-source peaks are 0.1343027 for full evolution, 0.1636293 for independent atoms, and 0.1447327 for the matched reference. Their sampled times are 1.1, 0.8, and 1.2, respectively.

The mean-field comparison uses $V_{\rm eff} = V(N-1)/N$ in the Bloch equations. For the dephased state, each initial coherence $(u_f + iv_f)e^{-i\theta}$ is evolved separately before averaging over θ. Evolving only the averaged vector $(n_f, 0, 0)$ would change the initial correlation structure under a nonlinear closure. At the $N = 32$ center, the resulting mean-field coherence signal differs from full evolution by 1.117 in relative L^2 norm over $0 \le t \le 12$, with maximum absolute difference 0.1498.

The separate size control holds $\Delta = \Delta_*(32)$ and the fold seed fixed. Table S8 records both its peak and initial trace

distance $D_0 = \|\delta\rho_c\|_1/2$. The latter uses the multiplicity-summed blocks of Eq. (S1), without another multiplicity factor. Fixed single-atom preparation therefore does not mean fixed global trace distance.

TABLE S8. Coherence-source size control at fixed detuning and single-atom seed. The peak is the largest sampled $|s_N|$ over $0 \le t \le 12$.

N	Density peak	D_0
1	0.1636293	0.32814
2	0.1145397	0.42925
4	0.1293140	0.55907
8	0.1349900	0.68136
16	0.1349931	0.77071
24	0.1345022	0.80829
32	0.1343027	0.83076
40	0.1342664	0.84645
48	0.1343079	0.85816

B. Detuning derivatives without a fitted response curve

For a fixed initial source, define $y_j = \partial_\Delta^j x(t)$. Since $L = L_0 + \Delta L_1$, the exact tangent equations are

$$\dot{y}_0 = Ly_0, \qquad \dot{y}_1 = Ly_1 + L_1 y_0, \qquad \dot{y}_2 = Ly_2 + 2L_1 y_1, \tag{S34}$$

with initial data $(x, 0, 0)$. We exponentiate this block-triangular generator and contract each component with N_e/N. The center predicts the two sides through $s(\Delta_* + h) = s_0 + hs_1 + h^2 s_2/2 + O(h^3)$, with $h = \pm w_N$. No side-point curve is fitted, and every mode remains included.

Table S9 gives the largest relative L^2 prediction error over the two endpoints, normalized separately by each full endpoint signal. The high-branch maximum $|\partial_\Delta s|$ changes only from 0.001534 to 0.001545 between $N = 24$ and 48. The coherence-source values rise from 0.2964 to 0.4450. Neither follows the much larger growth of $1/w_N$ in this window. Small prediction errors diagnose smooth local response; they do not establish mean-field validity or experimental precision.

TABLE S9. Fixed-source crossing tests. Δs is the high-branch signal difference between $\Delta_* \pm w_N$. The last two columns give relative errors of the complete-generator quadratic prediction. All quantities use $\gamma = 1$.

N	w_N	$\max_t \lvert\Delta s_{\rm hi}\rvert$	Coherence error	High-branch error
24	0.0264910	8.1281×10^{-5}	2.32×10^{-4}	1.03×10^{-5}
32	0.00747047	2.2984×10^{-5}	7.54×10^{-6}	1.79×10^{-7}
40	0.00212925	6.5658×10^{-6}	2.34×10^{-7}	4.11×10^{-9}
48	0.000612960	1.8941×10^{-6}	7.11×10^{-9}	9.95×10^{-11}

C. Finite splitting within the two-pole contribution

Write $\lambda_\pm = -\kappa \pm g$, with $\kappa > g > 0$, and set $A = a_- + a_+$ and $B = g(a_+ - a_-)$. The exact pair contribution and its coalesced-shape comparison are

$$S_2(t) = e^{-\kappa t}\left[A\cosh(gt) + B\frac{\sinh(gt)}{g}\right], \qquad S_0(t) = e^{-\kappa t}(A + Bt). \tag{S35}$$

This comparison holds A, B, κ fixed algebraically; it does not vary a physical generator. The opposite-residue expansion has precedents [10]. Applying $\cosh u - 1 \le u^2 e^u/2$ and $\sinh u - u \le u^3 e^u/6$ for $u \ge 0$ gives

$$|S_2 - S_0| \le g^2 e^{-(\kappa-g)t} \left(\frac{|A|t^2}{2} + \frac{|B|t^3}{6} \right), \tag{S36}$$

$$\sup_{t\ge 0} |S_2 - S_0| \le g^2 \left[\frac{2|A|}{e^2(\kappa-g)^2} + \frac{9|B|}{2e^3(\kappa-g)^3} \right]. \tag{S37}$$

For the coherence source, the all-time bounds at $N = 24, 32, 40, 48$ are approximately 8.18×10^{-5}, 1.04×10^{-5}, 1.18×10^{-6}, and 1.29×10^{-7}. The pair can contribute a nonexponential transient while the additional shape from its small splitting is limited. This is neither an error bound for complete evolution nor a CPTP replacement or metrological bound.

D. Trace-contraction bounds for the complete signal

For a fixed Hermitian difference $x = \rho_a - \rho_b$, put $D_0 = \|x\|_1/2$ and $O = N_e/N$, so $0 \le O \le I$. At real detuning, $T_\Delta(t) = e^{tL_\Delta}$ is CPTP and contracts the trace norm of Hermitian operators. Centering N_e by $NI/2$ gives $\|[N_e, Y]\|_1 \le N\|Y\|_1$. For traceless Hermitian Y, $|\operatorname{Tr}(OY)| \le \|Y\|_1/2$. Inserting $L_1 = i[N_e, \cdot]$ in the Duhamel formula therefore yields

$$|\partial_\Delta s(t)| \le N t D_0, \qquad |\partial_\Delta^2 s(t)| \le N^2 t^2 D_0. \tag{S38}$$

The second derivative contains two time-ordered insertions whose total integration volume is t^2. Integrating these bounds in detuning gives

$$|s(t, \Delta + w) - s(t, \Delta - w)| \le 2Nwt D_0, \tag{S39}$$

$$|s(t, \Delta + w) + s(t, \Delta - w) - 2s(t, \Delta)| \le N^2 w^2 t^2 D_0. \tag{S40}$$

These elementary estimates concern complete signals, independent of spectral truncation. They are loose for the present data: the largest endpoint and curvature ratios are below 0.0102 and 0.000319, respectively. If $N w_N \to 0$ were independently established, the first inequality would suppress fixed-time finite-amplitude features across that width. The finite-size fits do not prove this premise. Detuning-dependent preparations introduce additional source derivatives, so Eq. (S40) does not apply to the original varying-preparation scan without those terms.

E. Independent checks and retained failures

Explicit full-spin evolution at $N = 1, 4$ agrees with the corresponding PI density responses to less than 1.91×10^{-13}. The $N = 4$, $V = 0$ PI calculation agrees with the one-atom baseline to less than 4.21×10^{-12}. At $N = 32$, independent DOP853 integration checks both full transients to less than 4.04×10^{-14}. Five-point detuning differences from a separate $N = 2$ integration check Eq. (S34); at step 0.002, first- and second-derivative differences are below 8.83×10^{-12} and 3.50×10^{-11}.

The mean-field phase integral uses 64 samples, checked against 128. One outer point, $N = 48$ and $\delta = +0.06$, failed the 2×10^{-7} threshold with difference 4.81×10^{-7}. Its 256-phase refinement agrees with 128 phases to 1.04×10^{-11}. The failed record and original curve remain stored; the corrected curve is stored separately. The center and $\pm w_N$ conclusions do not depend on this refinement.

Subspace continuation takes eight steps per size from the center to $\pm 4w_N$, giving 32 points. Among each solver's returned candidates, neighboring subspace overlaps uniquely identify the same real pair; saved scan eigenvalues agree to less than 4.34×10^{-13}. Fourteen $N = 48$ candidate solves failed the 10^{-9} residual threshold and were retained before successful retries with a different shift. This is a candidate-set identity check, not global spectral isolation. Coordinate principal angles are not physical state fidelities. The full records, source hashes, rejected attempts, and successful refinements are in `data/physical_effects/`.

XI. STATIONARY POTENTIALS AND THE SCOPE OF THE SPLITTING EXPONENT

The ordinary Liouvillian gap is $\Gamma_{\text{diss}} = -\max_{\lambda \ne 0} \operatorname{Re}\lambda$ in the relevant symmetry sector. Our half-gap $g_N = (\lambda_+ - \lambda_-)/2$ instead compares two nonzero real eigenvalues. A prediction for Γ_{diss} therefore does not automatically predict g_N.

The exact steady-state construction of Roberts and Clerk [4] supplies analytical information for the model family containing Eq. (S3). Noachtar and Clerk [5] use hidden time-reversal symmetry and a particular steady-state purification to define a barrier. Their Conjecture 1 relates this barrier to the leading exponential scaling of Γ_{diss}, with numerical tests in selected examples. The purification potential is not an arbitrary excitation-number marginal, and the conjecture is not a theorem for the finite-rate doublet considered here. Their spin-model tests include collective decay; we do not claim they verify our purely local-decay parameter point.

Accordingly, comparing a finite-window fit for g_N with a stationary marginal rate function $\Phi(n_f)$ is descriptive. It supplies neither a derivation of the relevant boundary conditions nor a relation between the two spectral observables. The main text's Schur-root and coupling-product results remain finite-size identities and numerical checks, rather than an independent action prediction.

XII. REPRODUCTION AND EVIDENCE SCOPE

The source bundle provides commands, scripts, JSON records, arrays, and a SHA256 manifest. The environment used Python 3.13.2, NumPy 2.2.6, SciPy 1.16.3, and QuTiP 5.3.1, with one linear-algebra thread. All calculations use `biorthogonal_generator.py`. The stored $N = 24$ coherence sample predates local geometric bracketing, and all four original root files predate the final branch-discontinuity guards. All eight roots were subsequently back-substituted with the final verifier; raw samples and source hashes preserve this provenance. Earlier quasipotential tests and unconverged instanton trajectories are not used as quantitative evidence here.

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